Recommended Reading

7.1-7.4

Lecture Notes

Pre-Lecture Notes

Work

Work is generally defined as the integral of the path of a force with respect to it's displacement. To begin, we can define the definition of an infinitesimally small amount of work:

$$dW = \vec{F} \cdot d\vec{r} = |\vec{F}||d\vec{r}| \cos\theta$$

Note, $\vec{r}$ is the terms of our relative path length, and $\vec{F}$ is the terms of force applied onto our object. Work is generally defined as a term of $[N \cdot m] = [J]$.

$$W_{AB} = \int_{\text{path AB}} \vec{F} \cdot d\vec{r}$$

Where $\vec{F}$ could vary along the path, and where $d\vec{r}$ can also vary along the general path length.

Constant Forces & Work

Note that if $\vec{F} = \bar{F} = F$, that is force is some constant value; the above integral evaluates as the following:

$$\begin{aligned} W_{AB} &= \int_{\text{Path AB}} \vec{F} \cdot d\vec{r} = \int_{\text{Path AB}} dW \\ &= \vec{F} \int_{\text{Path AB}} d\vec{r} \\ &= \vec{F} \cdot (\vec{r}_B - \vec{r}_A) \\ &= |\vec{F}||\vec{r}_b - \vec{r}_a| \cos\theta \end{aligned}$$

With $\vec{r}_b - \vec{r}_a$ of the path of AB, this generally approximates the vector $\vec{d}$; therefore

$$= |\vec{F}||\vec{d}| \cos\theta$$

Or, without vector notation:

$$W = Fd\cos\theta$$

Non-Constant Forces & Work

In general for non-constant forces which vary over some distance; it may help to show them as a function of their components. Recall that we defined force in the direction of it's path as the following:

$$dW = \vec{F} \cdot d\vec{r}$$

If we take this to it's component form; then:

$$dW = F_x dx + F_y dy + F_z dz$$

Where F_x, F_y, F_z are the relative components of some force vector F.
Note then that if we are given the formula for some work; we can sub in and differentiate corresponding statements of $x, y, z \to dx, dy, dz$ to sub into this equation. We normally want to get these all to some evaluable term e.g. dx to take the definite integral of.

Hooke's Law & Elastic Energy

Hooke's Law is defined as the following:

$$F_E = -kx$$

Where k is some material component, and x is some displacement from the baseline. Note that this is a linear force (so we can use our simplified integral). If we wanted to solve for the linear elastic energy present in a spring; recall our definition from above:

$$W_{AB} = \int_{\text{path AB}} \vec{F} \cdot d\vec{r}$$

In this case, $\vec{F} = -kx$, and $d\vec{r}$ is equivalent to the path length, or *displacement* experienced by our spring, some term dx, therefore:

$$\begin{aligned} W_E &= \int_A^B -kx \; dx \\ &= -k \int_A^B x \; dx \\ &= -k \left[\frac{1}{2} x_b^2 - \frac{1}{2} x_a^2 \right] \\ W_E &= -\frac{k}{2} [x_b^2 - x_a^2] \end{aligned}$$

Kinetic Energy

Kinetic Energy is generally defined as the following:

$$K = \frac{1}{2} m v^2$$

> Note that K can sometimes be notated as E_K; this is in terms of Joules. Note that $p = mv \to v = \frac{p}{m}$, which means that this can be rewritten as:
>
> $$K = \frac{mp^2}{2m^2} = \frac{p^2}{2m}$$

Note however that v, like other values, are based off a certain reference frame. Refer to 6 — Rotational Kinematics and Newton's First Law (PHY180-7) for a brief review on this.

Work-Energy Theorem

The Work-Energy Theorem states that the Net Work of a system is equivalent to a change in it's kinetic energy. Let us retake the definition of work to show how this is true:

$$\begin{aligned} dW &= \vec{F} \cdot d\vec{r} && \text{Note, } \vec{F} = m\frac{d\vec{v}}{dt} \\ &= m\frac{d\vec{v}}{dt} \cdot d\vec{r} \\ &= m d\vec{v} \cdot \frac{d\vec{r}}{dt} && \text{Note, } \frac{d\vec{r}}{dt} = d\vec{v} \\ &= m d\vec{v} \cdot d\vec{v} && m d\vec{v} \to m\vec{v} \\ &= m\vec{v} \cdot d\vec{v} \end{aligned}$$

I personally (Aries) am not that familiar yet with integration of vectors; we can break this up into it's component form:

$$
\begin{aligned}
dW &= mv_x dv_x + mv_y dv_y + mv_z dv_z = m\left[v_x dv_x + v_y dv_y + v_z dv_z\right] \\
W_{net} = \int_A^B dW &= \int_A^B m\left[v_x dv_x + v_y dv_y + v_z dv_z\right] \\
&= \left[\frac{m}{2}v_x^2 + \frac{m}{2}v_y^2 + \frac{m}{2}v_x^2\right]_A^B \\
&= \left[\frac{m}{2}v_x^2 + \frac{m}{2}v_y^2 + \frac{m}{2}v_z^2\right]_A^B = [\frac{m}{2}v^2]_A^B \qquad\qquad v_x^2 + v_y^2 + v_z^2 = v^2 \\
W_{net} &= K_B - K_A = \Delta K
\end{aligned}
$$

What this means is that for instance, if Work is *negative*, then the kinetic energy of our object will proportionally decrease.

Power

Power is generally defined as some term of Energy over time, that is:

$$P_{av} = \frac{W}{t}$$

This is often notated in the term of a W (Watt), where $1W = \frac{1J}{s}$.

Instantaneous power is defined as the following:

$$P = \frac{dW}{dt}$$

Note that this can also be expressed as:

$$
\begin{aligned}
P &= \frac{dW}{dt} \qquad dW = \vec{F} \cdot d\vec{v} \\
&= \frac{\vec{F} \cdot d\vec{r}}{dt} \\
&= \vec{F} \cdot \frac{d\vec{r}}{dt} \\
&= \vec{F} \cdot \vec{v}
\end{aligned}
$$

Note that if P is constant for some time interval dt, then we can say that:

$$
\begin{aligned}
Pdt &= dW \\
\int Pdt &= \int dW \\
Pt &= W
\end{aligned}
$$

That is to say, that if P is constant, the net work is defined as the area under the Power curve with respects to time.